STAT C205A: Probability Theory

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We want to know when $\xi, \xi_1, \xi_2, \dots$ in L^1 with $\xi_n \xrightarrow{d} \xi$ implies $\mathbb{E}[\xi_n] \to \mathbb{E}[\xi]$.

Lemma 19.1: Uniform Integrability

Random variables ξ_t, $t \in T$ are uniformly integrable iff $\sup_t \mathbb{E}\,|\xi_t| < \infty$ and

$$\lim_{PA \to 0} \sup_{t \in T} \mathbb{E}[|\xi_t|\,; A] = 0$$

that is, for all $\epsilon > 0$, there exists $\delta > 0$ such that

$$P(A) < \delta \implies \mathbb{E}[|\xi_t|\,; A] < \epsilon \quad \forall t \in T$$

Proof: If ξ_t is uniformly integrable, then as

$$\mathbb{E}[|\xi_t|\,; A] \le rPA + \mathbb{E}[|\xi_t|\,; |\xi_t| > r], \quad r > 0$$

taking $PA \to 0$ and $r \to \infty$ at appropriate rates, we can get $\mathbb{E}[|\xi_t|\,; A] < \epsilon$ for all t, for any $\epsilon > 0$.

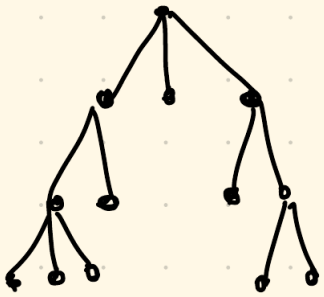
For boundedness, let $A = \Omega$ and take r sufficiently large.

Conversely, let ξ_t be L^1-bounded and satisfy the above condition. By Chebyshev,

$$\sup_t P\{|\xi_t| > r\} \le r^{-1} \sup_t \mathbb{E}\,|\xi_t| \to 0$$

as $r \to \infty$. Thus, if we set $A = \{|\xi_t| > r\}$, we see $PA \to 0$ uniformly as $r \to \infty$, so $\mathbb{E}[|\xi_t|\,; |\xi_t| > r] \to 0$. ■

Example: Critical Galton-Watson Branching Process



Clearly, $\mathbb{E}[\#\text{ offspring in } n^{\text{th}} \text{ generation}] = m^n$, where m is the expected number of offspring of a single person.

If $m > 1 \implies$ positive probability the tree never dies out.

If $m < 1 \implies$ dies out with probability 1.

If $m = 1 \implies$ dies out almost surely!

- $\mathbb{E}[Z_n] = 1$, but $\mathbb{E}[Z_n] \not\to 0 = \mathbb{E}[\lim_n Z_n]$

Lemma 19.2: Convergence of Means

Let $\xi, \xi_1, \xi_2, \ldots$ be $\mathbb{R}_+$-valued random variables with $\xi_n \xrightarrow{d} \xi$. Then,

(i) $\mathbb{E}\xi \le \liminf_n \mathbb{E}\xi_n$

(ii) $\mathbb{E}\xi_n \to \mathbb{E}\xi < \infty$ iff the ξ_n are uniformly integrable; that is,

$$\lim_{r\to\infty} \limsup_{n\to\infty} \mathbb{E}[|\xi_n| \,;\, |\xi_n| > r] = 0$$

Proof: **(i)** For any $r > 0$, $x \mapsto x \wedge r$ is bounded and continuous on $\mathbb{R}_+$, so

$$\liminf_{n\to\infty} \mathbb{E}[\xi_n] \ge \lim_{n\to\infty} \mathbb{E}[\xi_n \wedge r] = \mathbb{E}[\xi \wedge r]$$

as $\xi_n \to \xi$ in distribution. Letting $r \to \infty$, we see

$$\liminf_{n\to\infty} \mathbb{E}[\xi_n] \ge \mathbb{E}[\xi]$$

where we can turn $\liminf$ to $\lim$ as we know the limit exists.

(ii) ($\Leftarrow$) Assume uniform integrability. Then,

$$\mathbb{E}[\xi] \le \liminf_n \mathbb{E}[\xi_n] < \infty$$

(since they are L^1-bounded by above). Then, for any $r > 0$,

$$\begin{aligned}|\mathbb{E}[\xi_n] - \mathbb{E}[\xi]| \leq |\mathbb{E}[\xi_n] - \mathbb{E}[\xi_n \wedge r]| + |\mathbb{E}[\xi_n \wedge r] - \mathbb{E}[\xi \wedge r]| \\ + |\mathbb{E}[\xi \wedge r] - \mathbb{E}[\xi]|\end{aligned}$$

Now, $(x - x \wedge r) \leq x1\{x > r\}$, so as $\mathbb{E}[\xi_n; \xi_n > r]$ (uniform integrability) and $\mathbb{E}[\xi; \xi > r]$ (DCT) go to zero,

$$|\mathbb{E}[\xi_n] - \mathbb{E}[\xi_n \wedge r]| \to 0 \qquad |\mathbb{E}[\xi \wedge r] - \mathbb{E}[\xi]| \to 0$$

and by convergence in distribution,

$$|\mathbb{E}[\xi_n \wedge r] - \mathbb{E}[\xi \wedge r]| \to 0$$

so taking $n \to \infty$ and $r \to \infty$, we get $\mathbb{E}[\xi_n] \to \mathbb{E}[\xi]$.

($\Rightarrow$) Assume $\mathbb{E}\xi_n \to \mathbb{E}\xi < \infty$. Fix $r > 0$. Then, by continuity,

$$\mathbb{E}[\xi_n; \xi_n > r] \leq \mathbb{E}[\xi_n - \xi_n \wedge (r - \xi_n)_+] \to \mathbb{E}[\xi - \xi \wedge (r - \xi)_+]$$

since $x1\{x > r\} + x \wedge (r - x)_+ \leq x$. Then, as $x \wedge (r - x)_+ \uparrow x$ as $r \to \infty$, the RHS goes to 0 by DCT, so uniform integrability holds.

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